

Problem Set — Weeks 5–6

Artificial Intelligence and Ethics · VCE Algorithmics (HESS) · Unit 4 AOS3

Sections A–C correspond to Topics 15–17. Attempt each after its lesson.

Name: _____

Due: Tuesday 25 August

Note: this is the last problem set before the **SAC on Friday 28 August**. Sections A and C are written in SAC register — extended response, marks for structure as much as content.

Section A — Turing Test, Chinese Room, strong and weak AI (Topic 15)

A1. Distinguish **strong AI** from **weak AI**, one sentence each.

A2. Describe the setup of the Turing Test. Your answer must identify the participants, the communication channel, and the pass condition.

A3. State **two** limitations of the Turing Test as a measure of machine intelligence. For each, explain *why* it matters — a limitation without a consequence earns half marks.

A4. Describe Searle's Chinese Room Argument and explain how it is used to argue against the possibility of strong AI.

A5. State the **Systems Reply** to the Chinese Room, and give Searle's response to it.

A6. *Exam-style.* A company builds a system that inspects photographs of produce and grades them by ripeness more accurately than its human graders. The company claims this is evidence of artificial intelligence.

Explain whether the system would be considered artificial intelligence, referring to **two different conceptions** of artificial intelligence.

Section B — Machine learning and neural networks

(Topic 16)

B1. Describe a motivation for using a neural network rather than a traditional designed algorithm for classifying photographs.

B2. A single output node a receives three inputs i_0, i_1, i_2 with weights $X, 0.5$ and -3 respectively. The bias is 0. The inputs are set to $i_0 = 0.5, i_1 = 2, i_2 = 1$.

Find the value of X that makes the weighted sum at a equal to zero. Show your working.

B3. A small network has two inputs, two hidden neurons and one output, all with a step activation (fires if the weighted sum is > 0).

Hidden neuron h_1 : weights 0.6 and -0.4 , bias 0.2.

Hidden neuron h_2 : weights -0.5 and 0.9, bias 0.3.

Output neuron: weight 0.7 from h_1 , weight -0.8 from h_2 , bias 0.1.

For inputs $x_1 = 1, x_2 = 0$, compute the network's output. Show every weighted sum.

B4. State the SVM's optimisation objective, and explain why maximising it improves classification of *new* data.

B5. Points inside a circle cannot be separated from points outside it by any straight line in two dimensions. Explain how adding a dimension solves this, naming a suitable new feature.

B6. A model scores 99% accuracy on its training data and 62% on new data. Diagnose the problem and give two ways to address it.

B7. A student trains a network on 100 photographs, then tests it by renaming those same photographs "Photo 1", "Photo 2" and so on and running the model on the renamed files. Describe **two** problems with this approach.

Section C — Ethics of data-driven algorithms

(Topic 17)

C1. Name the four dimensions of AI ethics covered in this course, defining each in a clause.

C2. Distinguish **transparency** from **accountability**, and give an example of a system that has one without the other.

C3. Name the four mechanisms by which bias enters a data-driven system, with a one-line example of each.

C4. Two independent audits of the same classifier reach opposite conclusions about whether it is fair, and neither has made an error. Explain how this is possible.

C5. Exam-style. A community proposes that 1000 local residents contribute soil, climate and crop-yield data, which will be used to train a neural network. They hope the model will predict and explain successful crop production, and intend to sell it to other communities for use in their areas.

Describe one **technical** risk and one **ethical** risk with this approach.

C6. Four-move paragraph. A state government deploys a model that allocates public housing applicants to priority bands.

Write a single paragraph identifying one ethical issue, using the four moves: **stakeholder, harm, mechanism, mitigation**. Your mitigation must state what it costs.

Section D — SAC preparation

D1. The SAC covers all of AOS2 and AOS3. For each pair below, write one sentence that would prevent you confusing them under exam pressure.

- a.) intractable vs undecidable
- b.) NP-hard vs NP-complete
- c.) hill climbing vs simulated annealing
- d.) strong AI vs weak AI
- e.) transparency vs accountability
- f.) overfitting vs underfitting
